

- 5 (a) Many processes and actions in plants and animals are due to the movement of ions.

Table 5.1 lists several ions, the direction of movement of each ion and the action resulting from that movement.

Complete Table 5.1.

Table 5.1

ion	direction of movement of ion	action resulting from movement of ion
Na ⁺	from lumen of proximal convoluted tubule into proximal convoluted tubule cell	
.....	into guard cell	opening of stoma(ta)
.....	into presynaptic knob	depolarisation of presynaptic membrane
.....	into presynaptic knob	release of acetylcholine into synapse
H ⁺	from intermembrane space into matrix of mitochondrion	
Ca ²⁺	from sarcoplasmic reticulum to cytoplasm of muscle fibre	

(b) The Venus fly trap is a plant that is able to capture and digest insects. It has modified leaves, which have sensory hairs that respond to touch. When an insect comes into contact with the hairs, receptor potentials are generated.

(i) Name the ion that moves into the cells at the base of the sensory hairs to generate receptor potentials.

..... [1]

(ii) If two or more of these hairs are stimulated within a period of 20–35 seconds, action potentials are generated, causing the leaf to close quickly and trap the insect.

Suggest why it is beneficial to the plant for stimulation of two or more hairs to be necessary before the leaf will close.

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..... [1]

(iii) The trapped insect is digested by enzymes released from the leaf cells.

Name the mechanism by which the enzymes are released.

..... [1]

(iv) Suggest why Venus fly trap plants need to capture insects.

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..... [1]

[Total: 10]